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A distributed approach for lidar-based relative state estimation of multi-UAV in GPS-denied environments

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Abstract In this paper, we present a distributed framework for the lidar-based relative state estimator which achieves highly accurate, real-time trajectory estimation of multiple Unmanned Aerial Vehicles (UAVs) in GPS-denied environments. The system builds atop a factor graph, and only on-board sensors and computing power are utilized. Benefiting from the keyframe strategy, each UAV performs relative state estimation individually and broadcasts very partial information without exchanging raw data. The complete system runs in real-time and is evaluated with three experiments in different environments. Experimental results show that the proposed distributed approach offers comparable performance with a centralized method in terms of accuracy and real-time performance. The flight test demonstrates that the proposed relative state estimation framework is able to be used for aggressive flights over 5 m/s.

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1. Introduction

The deployment of the collaboration of UAVs has the potential to enable surveillance, search, information gathering, and more efficient coverage and monitoring of vast areas.^{1–6} Both

of these works rely on the availability of ground truth measurements. But external sensors, such as GPS,^{7,8} motion capture system,^{9,10} Ultra-Wideband (UWB),^{11,12} is unreliable in dynamic, partially-known environments. Therefore, it is necessary to develop a state estimation system that only relies on onboard sensing and computation, which is able to provide both state estimation and relative state estimation of a multi-UAV system. This paper provides a detailed description of a distributed approach for lidar-based relative state estimation which can be adopted by the multi-UAV systems in GPS-denied environments.

Robust state estimation is one of the key requirements for safe flight. For real-world flight, it is common to encounter

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complete darkness and low-texture environment, which may lead to the degradation of vision-based methods. Benefit from direct depth measurements, which is not affected by scene illumination and texture changes, lidar-based methods^{13,14} become a viable state estimator. However, lidar-based state estimator may also degrade in some geometric self-similarity environments. The fusion with an Inertial Measurement Unit (IMU) provides robustness and viable solution. Nevertheless, lidar-inertial odometry^{15,16} is only capable of providing the state estimations in local frames. Together with the drift issue, it is apparent that a simple one-time alignment of the coordinate systems of the individual UAVs before take-off is not sufficient during formation flying. It is intuitive to perform detection for other UAVs from the point cloud captured by the on-board lidar so as to obtain relative state estimations. Our goal is to design a distributed framework that allows each UAV to perform relative state estimation individually with minimal knowledge of the information of the teammates and sharing information with others.

1.1. Related work

Multi-robot Simultaneous Localization and Mapping (SLAM) has gained recent attention for the increased viability and accessibility of multi-UAV systems. In recent years, it is shown that researches on multi-UAV systems are more likely to focus on planning and control,^{17–20} and the relative state estimation is always leaped by adopting external positioning systems. Related work of relative state estimation investigates different sensing modalities, including vision,²¹ UWB,¹¹ and lidar.²²

Several works have adopted the vision-based methods for relative localization. In Ref.²³, a pattern detection-based visual relative localization approach is proposed. In the system, each UAV is equipped with a circle marker for detection purpose, which can be quite redundant. In Ref.²⁴, an ultraviolet light-based onboard relative localization method is presented, this method relies on additional UltraViolet Direction and Ranging (UVDAR) sensor and active marker for tracking purpose. However, due to the limitation of sensor Field-of-View (FOV), this method can only be applied to directed leader–follower-for-mation control. In Ref.²⁵, an inter-robot loop closures based collaborative localization method is running on a centralized ground station. The authors of Ref. 21 extend the method in Ref.²⁵ to the distributed system with an outlier rejection mechanism. Nonetheless, all the systems may encounter potential failure or drift when the points of interest are out of the FOV of the cameras, which limits the application in leader–follower formation.

UWB based ranging measurements can assist in relative state estimation. In Ref. 11, an onboard UWB ranging and communication network is adopted to sense the inter-UAV distance and exchange information for relative localization estimation in 2D spaces. In Ref.²⁶, a state estimator for both single and multi-robot systems is presented. However, this estimator requires a UWB anchor, which limits the application scenario. The method proposed in Ref.²⁷ represents a good solution for decentralized relative state estimation of the aerial swarm. However, solely relying on the vision for localization and mapping can be challenging as cameras are directional and sensitive to illumination changes and environmental conditions.

3D lidar is a viable sensing choice to map complex unstructured environments, which provides a 360° horizontal FOV and high sampling rate, and does not rely on external light sources. The authors of Ref.²⁸ present a strategy for collaborative mapping of an earthquake-damaged building via ground and aerial robots. In Ref.²⁹, an integrated autonomous 3D mapping system for multi-robot exploration is proposed. However, the map-merging strategies proposed in Ref.²⁸ and Ref.²⁹ are both performed offline. In Ref. 30 and Ref. 31, the authors present an online localization and mapping system for multiple ground robots equipped with 3D lidars, which is available open-source and easy to compare. In Ref.²², a lidar-based multi-robot SLAM system is presented, which developed in the context of the DARPA Subterranean Challenge. Nonetheless, both methods proposed in Ref. 30, Ref. 31, and Ref. 22 are centralized systems, all of them rely on a central server with potentially big computational capacity. The method proposed in Ref.³² represents a good solution for lidar-based decentralized relative state estimation for ground robots. To the best of our knowledge, most of the works related to lidar-based relative state estimation technology are centralized system or demonstrated with ground robots. This is the first time a distributed lidar-based relative state estimation framework is demonstrated on a multi-UAV system during a real-world aggressive flight.

1.2. Contribution

A lidar-based on-board real-time relative state estimation system is proposed in this paper, which can be adopted by multi-UAV systems. The main contributions are as follows:

- (1) A distributed relative state estimation framework is developed to build atop a factor graph, that is suitable for the multi-UAV systems. Only onboard sensors and computing power, including lidars, inertial measurement units, as well as Intel NUC, are utilized. No external off-board sensors or computing power are required;
- (2) A keyframe-based lidar-inertial odometry and inter-UAV loop closures detection method is proposed with an outlier rejection mechanism. Each UAV performs relative state estimation individually and broadcasts very partial information, including keyframes and odometry information, without exchanging raw data;
- (3) The proposed framework is validated with an extensive experimental evaluation including indoor test, outdoor test, and aggressive flight test in the real world.

2. System overview

Fig. 1 shows the architecture of our lidar-based relative state estimation framework. The goal of each UAV is to estimate its own trajectory and relative pose of the observed UAV. The modules of each UAV can be categorized into 6 main parts: on-board sensor, lidar-inertial odometry, keyframe selection, distributed place recognition, relative pose-graph optimization, map management unit. The on-board sensor consists of 3D lidar and IMU, which is a popular choice as a sensing payload since they provide point clouds, acceleration, and angular velocity to enable UAV state estimation in

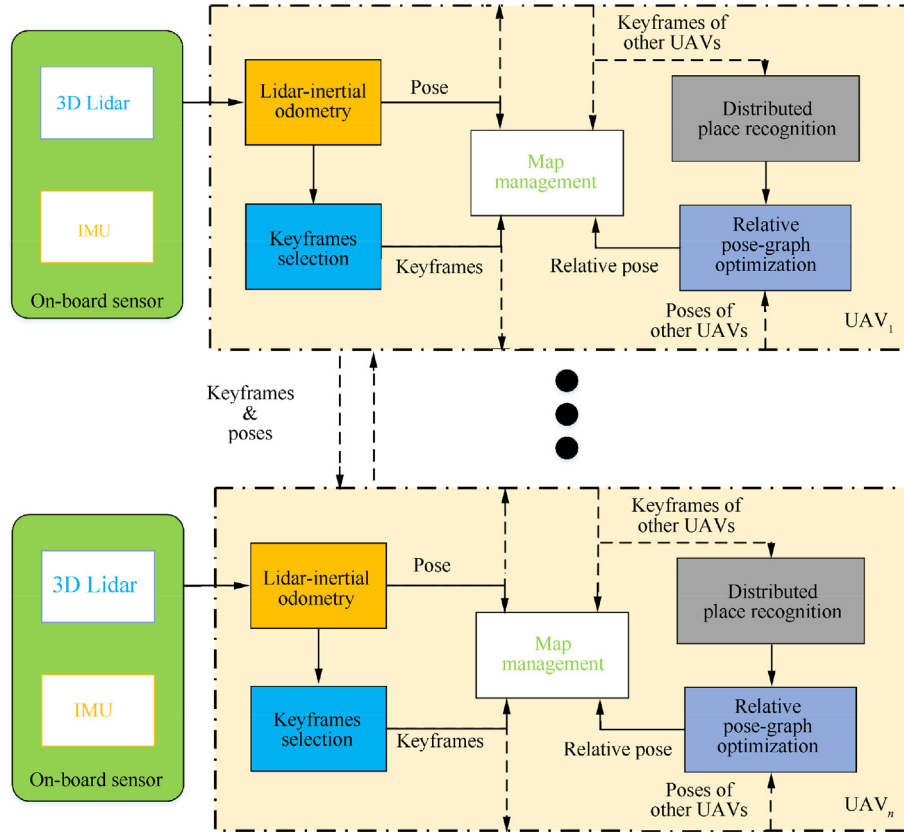


Fig. 1 Overview of the proposed relative state estimation architecture.

darkness and low-texture environments. Lidar-inertial odometry unit seeks to estimate the state of UAV and its trajectory using the observation of these sensor data. As done in other lidar odometry pipelines,¹⁵ we also use a scan-to-map matching step, where the latest scan is matched against a keyframe-based map. The maps are maintained on each UAV using a local inertial frame, while information between the UAVs is exchanged in relative coordinates from one local inertial frame to the other. Note that, map management module maintains a keyframe-based map by forwarding newly arrived keyframes to the map. Maintaining the keyframe-based map not only achieves a balance between map density and memory consumption but also reduces communication bandwidth requirement. The distributed place recognition

module collects keyframes of other UAVs and establishes relative poses between the UAVs trajectories using the Iterative Closest Point (ICP) algorithm. The relative pose-graph optimization module computes relative poses using keyframes and poses that are observed at other UAVs by incremental relative pose-graph optimization.

2.1. Notations

To help the understanding of system framework, nomenclatures are defined in Table 1.

Table 1 Nomenclature.

$\widehat{(\cdot)}$	The estimated state	$F_{t_k}^q$	All planar features extracted at time t_k
$\bar{(\cdot)}$	The initial state	$l_i M_{t_k}$	Keyframe based map maintained on UAV _i at time t_k
$l_i(\cdot)$	State in UAV _i 's inertial frame	$d_{t_k}^e$	The distance between a feature and its edge patch correspondence at time t_k
$B_i(\cdot)$	State in UAV _i 's body frame	$d_{t_k}^p$	The distance between a feature and its planar patch correspondence at time t_k
$l_i T_{j,t_k}$	The relative pose of UAV _j in UAV _i 's inertial frame at time t_k	z_{t_k}	The odometry measurement at time t_k
F_{t_k}	All features extracted at time t_k	$l_i z_{j,t_k}$	The place recognition measurement describes a measurement of the relative pose between the UAV _j and UAV _i 's inertial frame at time t_k
$F_{t_k}^e$	All edge features extracted at time t_k	σ	Gaussian noise in the estimation process

2.2. Problem formulation

We consider a multi-UAV system that contains n UAVs (labeled consecutively from 1 to n), that each UAV $_i$ has an inertial frame I_i and a body-fixed frame B_i , for $i = 1, 2, \dots, n$. Each UAV is equipped with a 3D lidar and an IMU in order to run lidar-inertial odometry, in which the ego-motion state $T \in SE(3)$ is able to be estimated in its own inertial frame. Meanwhile, each UAV will broadcast certain information, including pose, and keyframes, to other UAVs. The relative state estimation problem can be represented as: For an arbitrary UAV $_i$, estimate the pose of each UAV $_j$ in UAV $_i$'s inertial frame I_i . This relative state estimation problem can be formulated as a Maximum a Posteriori (MAP) problem. With the assumption of a Gaussian noise model, the MAP inference for our problem is equivalent to solving a nonlinear least-squares problem.

$${}^I_i \hat{T}_{j,t_k} = \argmin ({}^I_i T_{j,t_k})^{-1} ({}^I_i \hat{T}_{j,t_k}) \quad (1)$$

where ${}^I_i \hat{T}_{j,t_k}$ is solved using a versatile relative pose-graph optimization and implemented in Georgia Tech Smoothing and Mapping (GTSAM) library.³³

3. Relative state estimation framework

In this section, a lidar-based relative state estimation framework is presented in a distributed fashion. The functionalities of the individual modules shown in Fig. 1 are described in the following.

3.1. Lidar-inertial odometry

The lidar-inertial odometry is in charge of using the raw sensor data to obtain pose measurements of a single UAV. Inspired by Ref.¹⁵, a keyframe strategy is proposed to reduce memory consumption. Compare to traditional two-stage odometry estimation, we use IMU prediction instead of scan-to-scan to reduce the computational cost, without reducing localization precision. As done in other lidar odometry pipelines,¹⁵ a scan-to-map matching step is utilized to produce an accurate pose estimate.

When a new lidar scan arrives, features are extracted by the algorithm proposed in Ref.¹³. The feature extraction unit extracts points on edge or plane by calculating the curvature of each point. Points with a large curvature value are classified as edge features. Similarly, a planar feature is categorized by a small curvature value. All feature points extracted at time t_k compose a feature point set ${}^B F_{t_k}$ which consists of a subset of edge points ${}^B F_{t_k}^e$ and planar points ${}^B F_{t_k}^p$. Considering the sparsity of point cloud in a single lidar scan, we store point cloud on map ${}^I M_{t_k}$, this can be done using the map management unit described in Section 3.5. For each edge point ${}^I f_{q,t_k}^e \in {}^I F_{t_k}^e$, its correspondence in ${}^I M_{t_{k-1}}$ is a line represented by two edge points ${}^I f_{u,t_{k-1}}^e$ and ${}^I f_{v,t_{k-1}}^e$. The distance between an edge feature and its correspondence can be defined as d_{odom}^e in Eq. (2). For each planar point ${}^I f_{q,t_k}^p \in {}^I F_{t_k}^p$, its correspondence in ${}^I M_{t_{k-1}}$ is a plane represented by planar points

${}^I f_{u,t_{k-1}}^p$, ${}^I f_{v,t_{k-1}}^p$ and ${}^I f_{w,t_{k-1}}^p$. The point-to-plane distance d_{odom}^p can be defined as Eq. (3).

$$d_{\text{odom},t_k}^e = \frac{\left| \left({}^I f_{q,t_k}^e - {}^I f_{u,t_{k-1}}^e \right) \times \left({}^I f_{q,t_k}^e - {}^I f_{v,t_{k-1}}^e \right) \right|}{\left| \left({}^I f_{u,t_{k-1}}^e - {}^I f_{v,t_{k-1}}^e \right) \right|} \quad (2)$$

$$d_{\text{odom},t_k}^p = \frac{\left| \left({}^I f_{q,t_k}^p - {}^I f_{u,t_{k-1}}^p \right) \times \left({}^I f_{q,t_k}^p - {}^I f_{v,t_{k-1}}^p \right) \right|}{\left| \left({}^I f_{u,t_{k-1}}^p - {}^I f_{v,t_{k-1}}^p \right) \times \left({}^I f_{u,t_{k-1}}^p - {}^I f_{w,t_{k-1}}^p \right) \right|} \quad (3)$$

where ${}^I f_{t_k}^e$ and ${}^I f_{t_k}^p$ are the edge and planar feature points in the inertial frame I that transformed by T_{t_k} . Therefore, both d_{odom,t_k}^e and d_{odom,t_k}^p can be expressed as functions of transformation $\hat{T}_{t_k}$.

$$f(\hat{T}_{t_k}) = \begin{bmatrix} d_{\text{odom},t_k}^e \\ d_{\text{odom},t_k}^p \end{bmatrix} \quad (4)$$

then $\hat{T}_{t_k}$ can be solved through nonlinear iterations by minimizing d_{odom,t_k}^e and d_{odom,t_k}^p toward zero.

$$\hat{T}_{t_k} = \argmin_{\hat{T}_{t_k}} f(\hat{T}_{t_k}) \quad (5)$$

Note that, the initial transformation of Eq. (5) is obtained by using the predicted UAV motion from IMU, rather than using scan-to-scan matching like LOAM.¹³ Meanwhile, IMU measurements are exploited to provide a prediction at a higher rate.

3.2. Keyframe selection

To improve the real-time performance of the lidar odometry, we adopt the concept of keyframe selection.¹⁵ The keyframe selection criteria can be represented as: If the change in UAV pose between the current frame and the latest keyframe is beyond a certain threshold, the current frame is treated as a new keyframe. The keyframe strategy provides a trade-off between map density and memory consumption that makes the global map maintained in the map management unit suitable for scan-to-map matching.

3.3. Distributed place recognition

The distributed place recognition unit produces noisy relative pose measurements of UAVs. For instance, the relative pose ${}^I_i T_j$ describes a measurement of the relative pose between the UAV $_j$ and UAV $_i$'s own inertial frame I_i . The distributed place recognition unit of UAV $_i$ targets to match the current place of UAV $_j$ with the previously visited places of UAV $_i$ from the sub-keyframes $\{{}^I_i F_{t_{k-n-m}}, \dots, {}^I_i F_{t_{k-n}}, \dots, {}^I_i F_{t_{k-n+m}}\} \in {}^I_i M_{t_{k-1}}$, where ${}^I_i F_{t_{k-n}}$ is one of the candidate target keyframe feature sets for place recognition. For UAV $_j$, features are extracted by the algorithm proposed in Section 3.1. When ${}^B_j F_{t_k}$ is a keyframe, UAV $_j$ will broadcast it to all the other UAVs. For a newly obtained feature point set ${}^B_j F_{t_k}$, UAV $_i$ first transform ${}^B_j f_{t_k} \in {}^B_j F_{t_k}$ from UAV $_j$'s body fixed frame B_j to UAV $_i$'s inertial frame I_i .

$${}^I_i f_{j,t_k} = {}^I_i T_{j,t_k}^{B_j} f_{t_k} \quad (6)$$

Then found the closest neighbor of ${}^I_i f_{j,t_k}$ in the sub-keyframes $\{{}^I_i F_{t_{k-n-m}}, \dots, {}^I_i F_{t_{k-n}}, \dots, {}^I_i F_{t_{k-n+m}}\}$ as the correspondence features. As described in Section 3.1, the distance between an edge feature ${}^I_i f_{j,q,t_k} \in {}^I_i F_{j,t_k}^e$ and its correspondence ${}^I_i f_{u,t_{k-1}}, {}^I_i f_{v,t_{k-1}} \in {}^I_i M_{t_{k-1}}^e$ can be defined as d_{RelPose}^e in Eq. (7). The distance between a planar feature ${}^I_i f_{j,q,t_k} \in {}^I_i F_{j,t_k}^p$ and its correspondence ${}^I_i f_{u,t_{k-1}}, {}^I_i f_{v,t_{k-1}}, {}^I_i f_{w,t_{k-1}} \in {}^I_i M_{t_{k-1}}^p$ can be defined as d_{RelPose}^p in Eq. (8).

$$d_{\text{RelPose},t_k}^e = \frac{\left| \left({}^I_i f_{j,q,t_k}^e - {}^I_i f_{u,t_{k-1}}^e \right) \times \left({}^I_i f_{j,q,t_k}^e - {}^I_i f_{v,t_{k-1}}^e \right) \right|}{\left| \left({}^I_i f_{u,t_{k-1}}^e - {}^I_i f_{v,t_{k-1}}^e \right) \right|} \quad (7)$$

$$d_{\text{RelPose},t_k}^p = \frac{\left| \begin{pmatrix} {}^I_i f_{j,q,t_k}^p - {}^I_i f_{u,t_{k-1}}^p \\ \left({}^I_i f_{u,t_{k-1}}^p - {}^I_i f_{v,t_{k-1}}^p \right) \times \left({}^I_i f_{u,t_{k-1}}^p - {}^I_i f_{w,t_{k-1}}^p \right) \end{pmatrix} \right|}{\left| \begin{pmatrix} {}^I_i f_{u,t_{k-1}}^p - {}^I_i f_{v,t_{k-1}}^p \\ \left({}^I_i f_{u,t_{k-1}}^p - {}^I_i f_{v,t_{k-1}}^p \right) \times \left({}^I_i f_{u,t_{k-1}}^p - {}^I_i f_{w,t_{k-1}}^p \right) \end{pmatrix} \right|} \quad (8)$$

Substituting Eq. (6) into Eqs. (7) and (8), both d_{RelPose,t_k}^e and d_{RelPose,t_k}^p can be expressed as functions of transformation ${}^I_i \hat{T}_{j,t_k}$.

$$\begin{cases} f({}^I_i \hat{T}_{j,t_k}) = \begin{bmatrix} d_{\text{RelPose},t_k}^e \\ d_{\text{RelPose},t_k}^p \end{bmatrix} \\ d_{\text{RelPose},t_k}^e = \frac{\left| \left({}^I_i \hat{T}_{j,t_k}^{B_j} f_{j,q,t_k} - {}^I_i f_{u,t_{k-1}}^e \right) \times \left({}^I_i \hat{T}_{j,t_k}^{B_j} f_{j,q,t_k} - {}^I_i f_{v,t_{k-1}}^e \right) \right|}{\left| \left({}^I_i f_{u,t_{k-1}}^e - {}^I_i f_{v,t_{k-1}}^e \right) \right|} \\ d_{\text{RelPose},t_k}^p = \frac{\left| \begin{pmatrix} {}^I_i \hat{T}_{j,t_k}^{B_j} f_{j,q,t_k}^p - {}^I_i f_{u,t_{k-1}}^p \\ \left({}^I_i f_{u,t_{k-1}}^p - {}^I_i f_{v,t_{k-1}}^p \right) \times \left({}^I_i f_{u,t_{k-1}}^p - {}^I_i f_{w,t_{k-1}}^p \right) \end{pmatrix} \right|}{\left| \begin{pmatrix} {}^I_i f_{u,t_{k-1}}^p - {}^I_i f_{v,t_{k-1}}^p \\ \left({}^I_i f_{u,t_{k-1}}^p - {}^I_i f_{v,t_{k-1}}^p \right) \times \left({}^I_i f_{u,t_{k-1}}^p - {}^I_i f_{w,t_{k-1}}^p \right) \end{pmatrix} \right|} \end{cases} \quad (9)$$

then ${}^I_i \hat{T}_{j,t_k}$ can be solved through nonlinear iterations by minimizing d_{RelPose,t_k}^e and d_{RelPose,t_k}^p toward zero.

$${}^I_i \hat{T}_{j,t_k} = \underset{{}^I_i \hat{T}_{j,t_k}}{\text{argmin}} f({}^I_i \hat{T}_{j,t_k}) \quad (10)$$

3.4. Relative pose-graph optimization

This module is in charge of estimating the UAVs' relative state given the odometry measurements from the lidar-inertial odometry unit and the relative pose measurements from the distributed place recognition unit. We use a factor graph to model the relative state estimation problem. The factor graph implements 2 different types of factors, i.e., odometry factor, and place recognition factor which include both inter-UAV associations and intra-UAV loop-closures. For instance, Fig. 2 shows the factor graph of the UAV_i's relative state estimator in a multi-UAV system with 2 UAVs (labeled with $i, j, i \neq j$).

3.4.1. Odometry factor

The lidar-inertial odometry data is used to compute the odometry factor between adjacent nodes in the graph. For a multi-UAV system, each UAV locally estimates its own pose and broadcasts it to other UAVs. The odometry measurement can be obtained and formulated as follow.

$$\mathbf{z}_{t_k} = \mathbf{z}_{t_{k-1}} (\Delta T_{t_k} + \sigma_{\text{odom}}) \quad (11)$$

where ΔT_{t_k} is the transformation from time t_{k-1} to t_k , which are defined as follow.

$$\Delta T_{t_k} \triangleq T_{t_{k-1}}^{-1} T_{t_k} \quad (12)$$

Note that, Eq. (12) does not contain frame transformations because ΔT_{t_k} is frame invariant. Eq. (12) can be rewritten as follow.

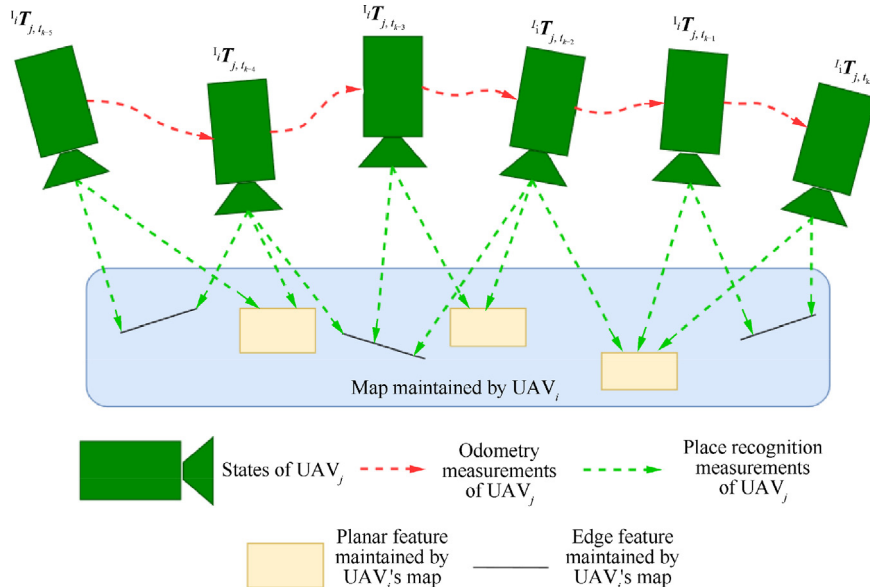


Fig. 2 Factor graph representation of relative state estimator.

$$\Delta \mathbf{T}_{t_k} \triangleq {}^{I_i} \mathbf{T}_{j,t_{k-1}}^{-1} {}^I \mathbf{T}_{j,t_k} \quad (13)$$

Substituting Eq. (13) into Eq. (11), the odometry measurements can be obtained and formulated by the displacements between consecutive keyframes with a Gaussian model.

$$\Delta \mathbf{z}_{t_k} \triangleq \mathbf{z}_{t_{k-1}}^{-1} \mathbf{z}_{t_k} = {}^{I_i} \mathbf{T}_{j,t_{k-1}}^{-1} {}^I \mathbf{T}_{j,t_k} + \sigma_{\text{odom}} \quad (14)$$

This intra-UAV measurements involves the poses of a single UAV at different time instants. Given the observation in Eq. (14), the residual from odometry can be defined as follow.

$$\mathbf{r}_{\text{odom}}({}^{I_i} \hat{\mathbf{T}}_{j,t_{k-1}}, {}^{I_i} \hat{\mathbf{T}}_{j,t_k}, \Delta \mathbf{z}_{t_k}) = \Delta \mathbf{z}_{t_k} {}^{I_i} \hat{\mathbf{T}}_{j,t_k}^{-1} {}^I \hat{\mathbf{T}}_{j,t_{k-1}} \quad (15)$$

3.4.2. Place recognition factor

To implement place recognition, loop closures detection takes a source keyframe point cloud set and a target sub-keyframes point cloud set as input. We compare the current keyframe with scans in the target point cloud within a given radius. Since compact lidars, such as the Velodyne Puck, have a relatively narrow vertical FOV (30°) and low vertical angular resolution (2.0°, 16 scanning channels), the points cloud is sparsity out of a certain range. In our tests, we choose this search distance for loop closures to be 15 m. The current keyframe is then registered against each of the nearby scans using ICP to obtain a transformation. For example, ${}^{I_i} \mathbf{F}_{j,t_k}$ is the current feature point set of UAV_j in UAV_i's inertial frame. We try to match ${}^{I_i} \mathbf{F}_{j,t_k}$ to the sub-keyframes $\{{}^{I_i} \mathbf{F}_{j,t_{k-n-m}}, \dots, {}^{I_i} \mathbf{F}_{j,t_{k-n}}, \dots, {}^{I_i} \mathbf{F}_{j,t_{k-n+m}}\}$, using the scan-matching method discussed in Section 3.3. We obtain the relative transformation $\Delta {}^{I_i} \mathbf{T}_{j,t_k}$ and add it as a correction to ${}^{I_i} \hat{\mathbf{T}}_{j,t_k}$.

$${}^{I_i} \hat{\mathbf{T}}_{j,t_k} = \Delta {}^{I_i} \hat{\mathbf{T}}_{j,t_k} {}^{I_i} \bar{\mathbf{T}}_{j,t_k} \quad (16)$$

where the initial transformation ${}^{I_i} \bar{\mathbf{T}}_{j,t_k}$ is obtained by adding the odometry factor to the graph. The place recognition result is also formulated with a Gaussian model.

$${}^{I_i} \mathbf{z}_{j,t_k} = {}^{I_i} \mathbf{T}_{j,t_k} + \sigma_{\text{RelPose}} \quad (17)$$

Given the observation in Eq. (17), the residual from place recognition can be defined as follow.

$$\mathbf{r}_{\text{RelPose}}({}^{I_i} \hat{\mathbf{T}}_{j,t_k}, {}^{I_i} \mathbf{z}_{j,t_k}) = {}^{I_i} \mathbf{z}_{j,t_k} {}^{I_i} \hat{\mathbf{T}}_{j,t_k}^{-1} \quad (18)$$

Before executing optimization, the distributed outlier rejection mechanism is adopted to check the quality of loop closures and remove outliers. This is done by checking the fitness score of each loop closure and reject loop closures with a score larger than a threshold (2.0 in our experiments), and then checking pairwise consistency between pairs of loop closures. Inspired by Ref.³⁴, the following metric is used to determine if two inter-robot loop closures ${}^{I_i} \mathbf{z}_{j,t_k}$ and ${}^{I_i} \mathbf{z}_{j,t_{k-n}}$ are pairwise consistent.

$$\left\| {}^{I_i} \mathbf{z}_{j,t_{k-n}} \prod_{t \in (t_{k-n}, t_k)} \Delta \mathbf{z}_t \mathbf{z}_{j,t_k}^{-1} \right\|_{\Sigma} \leq \gamma \quad (19)$$

The threshold γ can be determined from the quantile of the chi-squared distribution for a given probability level. Place recognition measurements that satisfy Eq. (19) are considered inlier and added to the factor graph.

3.4.3. Optimization

The factor graph of the relative state estimator is shown in Fig. 2. This relative state estimation problem can be formulated as a MAP problem. With the assumption of a Gaussian noise model in Eqs. (14) and (17), the MAP problem inference for our problem is equivalent to solving a nonlinear least-squares problem. For example, the optimization for relative pose estimation of UAV_j in the UAV_i's inertial frame is expressed as the following formulation.

$$\min_{{}^{I_i} \hat{\mathbf{T}}_{j,t_k}} \left\{ \left\| \mathbf{r}_{\text{odom}}({}^{I_i} \hat{\mathbf{T}}_{j,t_{k-1}}, {}^{I_i} \hat{\mathbf{T}}_{j,t_k}, \Delta \mathbf{z}_{t_k}) \right\|^2 + \left\| \mathbf{r}_{\text{RelPose}}({}^{I_i} \hat{\mathbf{T}}_{j,t_k}, {}^{I_i} \mathbf{z}_{j,t_k}) \right\|^2 \right\} \quad (20)$$

This nonlinear least-squares problem considers 2 factors, corresponding to the odometry factor (Section 3.4.1), and the place recognition factor (Section 3.4.2). Simply solving Eq. (20) can directly provide the result of the relative state estimation.

3.5. Map management

Each UAV_i ($i = 1, 2, \dots, n$) in the system is in charge of maintaining a target map, ${}^{I_i} \mathbf{M}_{t_k}$, in its own inertial frame and generating its current keyframe feature point set ${}^{B_j} \mathbf{F}_{t_k}$, which is, along with odometry factor, the only information communicated to other UAVs. Converting the raw point cloud into ${}^{B_j} \mathbf{F}_{t_k}$ induces a high level of compression which is a key advantage for reducing the required communication bandwidth in multi-UAV systems. For a newly obtained keyframe feature point set ${}^{B_j} \mathbf{F}_{j,t_k}$ of UAV_j, the map management unit of UAV_i first transforms ${}^{B_j} \mathbf{F}_{j,t_k}$ into the UAV_i's inertial frame by using Eq. (6). Then merges transformed keyframe feature point set ${}^{I_i} \mathbf{F}_{j,t_k}$ together into the target map ${}^{I_i} \mathbf{M}_{t_k}$. Finally, ${}^{I_i} \mathbf{M}_{t_k}$ is downsampled to eliminate the duplicated features that fall in the same voxel cell, the downsample resolutions for ${}^{I_i} \mathbf{M}_{t_k}^q$ and ${}^{I_i} \mathbf{M}_{t_k}^e$ are 0.2 m and 0.4 m in our experiments, where ${}^{I_i} \mathbf{M}_{t_k}^q$ and ${}^{I_i} \mathbf{M}_{t_k}^e$ are edge feature voxel map and planar feature voxel map, respectively.

4. Experiments

We now describe three experiments to analyze the relative state estimation framework stated in Section 3. The performance of the proposed method in different scenarios is compared with SegMap^{30,31} which successfully integrates different state-of-the-art modules that are advantageous for multi-robot systems with regards to closing loops, computational complexity, and communication bandwidth requirements. During experiments, the interconnection topology of the multi-UAV system is strong connectedness (every UAV is reachable from other UAVs) based on the multi-master network of Robot Operating System (ROS). In the first experiment, an indoor test is introduced to evaluate the accuracy of the proposed method in 2 UAVs scenario by it comparing with ground truth data obtained from the OptiTrack motion capture system. Considering that the limited working area (2.5 m × 7 m) of the motion capture system prevents the real flight experiments of multi-UAV, a full functional experiment with manual move-

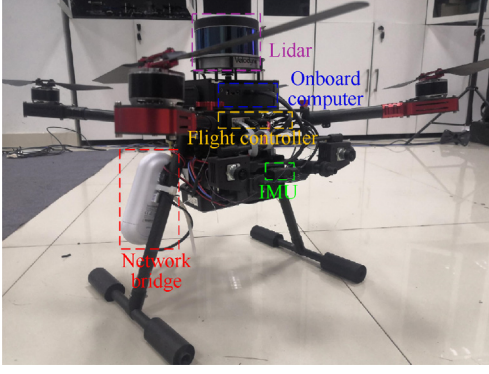


Fig. 3 One of the aerial platforms in the multi-UAV system.

ment of 2 UAVs is presented in Section 4.2 to evaluate the accuracy of the proposed method. Furthermore, the proposed method is also implemented on a real platform as shown in Fig. 3 for the outdoor experiments in Section 4.3 and Section 4.4. In the second experiment, a campus is chosen as the experiment area to evaluate the performance of the proposed method in a large-scale environment by checking the consistency with the Google Earth imagery. In the third experiment, a flight test is introduced to verify the validity and effectiveness of the proposed distributed relative state estimation method during aggressive flight.

4.1. Aerial platform

For the multi-UAV system, each UAV uses an identical hardware configuration. As shown in Fig. 3, one of the aerial platforms of the multi-UAV system is equipped with a Pixhawk4 flight controller, a Velodyne VLP-16 lidar, a MicroStrain 3DM-GX5-25 IMU, and an onboard computer (Intel i7-8550U CPU, 32 GB RAM). The raw data from the IMU at 1000 Hz and point clouds from the lidar at 10 Hz are fused together for the lidar-inertial odometry of individual UAVs, while the keyframes of lidar odometry are also utilized for relative state estimation. The attitude controller is performed by a widely used off-the-shelf Pixhawk4 attitude controller, while all other algorithms, including state estimation, trajectory planning, and position control, are running on the onboard computer. All algorithms are implemented in C++, and the ROS in Ubuntu 16.04 is used for communication between UAVs over a network bridge.

4.2. Indoor test

In this test, we focus on evaluating the accuracy of our relative state estimation framework in 2 UAV scenario (UAV₁ and UAV₂). During the data gathering, the user holds the UAV shown in Fig. 3 and walks quickly under the OptiTrack motion capture system. The relative trajectories from the proposed method and SegMap are shown in Fig. 4 and compared with

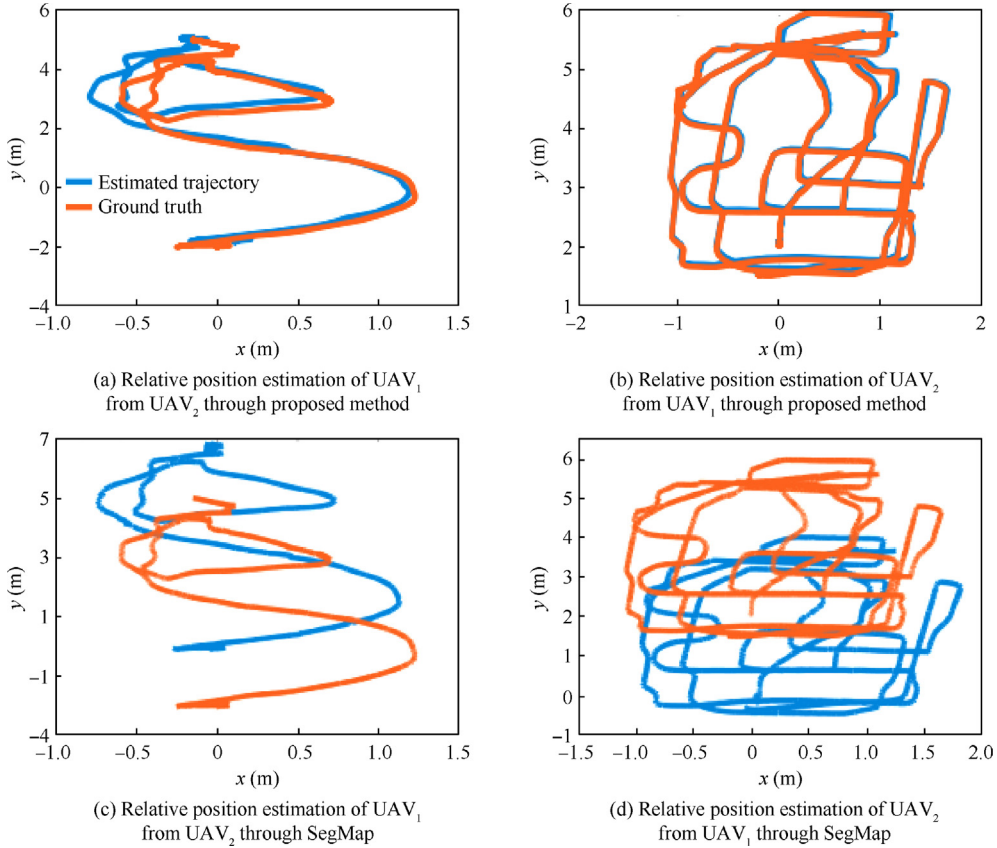


Fig. 4 Comparison of relative trajectories between different methods and ground truth in 2 UAV scenario.

Table 2 Comparison of relative state estimation with ground truth.

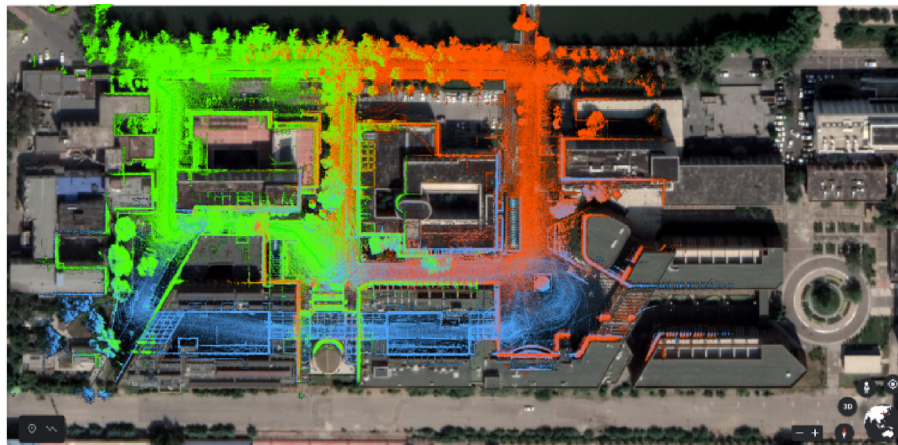
RMSE	${}^I_2 T_1$		${}^I_1 T_2$	
	Proposed method	SegMap	Proposed method	SegMap
$x(m)$	0.22	0.36	0.11	0.81
$y(m)$	0.18	2.26	0.13	2.30
$z(m)$	0.24	0.71	0.23	0.43
$\psi(^{\circ})$	1.02	1.23	3.26	5.35

the motion capture system introduced as ground truth. Due to the lack of typical semantic classes, such as vehicles and trees, in indoor environments, SegMap fails to initialize properly and produce meaningful results. The results of the developed method, shown in Fig. 4(a) and (b), are consistent with the ground truth. Table 2 shows the Root Mean Square Error (RMSE) results of the proposed method and SegMap. The results show that the proposed method outperforms SegMap and RMSE can reach 20 cm level in the indoor test environment.

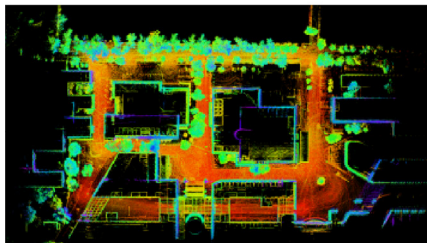
4.3. Outdoor test

To evaluate the performance of the proposed method in a large-scale environment ($275\text{ m} \times 160\text{ m}$), outdoor tests are performed with 3 UAVs running the proposed relative state estimation method. The 3 UAVs start at different locations, traverse different paths, and return to their starting points.

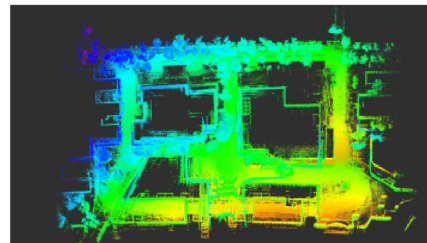
Fig. 5 shows the mapping results of the developed distributed method and the centralized SegMap method. The map of the proposed distributed method and a centralized method, SegMap, are shown in Fig. 5(b) and (c), respectively. From the results, it can be observed that the map of SegMap shows numerous blurry structures due to the poor relative state estimation results, and the map obtained by the developed method is well-aligned with Google Earth imagery. The keyframes of UAV₁, UAV₂, and UAV₃ are colored in blue, green, and red, respectively. The map management unit of UAV₁ uses ${}^I_1 T_k$ from its lidar-inertial odometry and ${}^I_1 T_{2,t_k}$, ${}^I_1 T_{3,t_k}$ from its relative state estimator to transform keyframes into UAV₁'s inertial frame I_1 , and the mapping result is shown in Fig. 5(b). The relative translational error of relative state estimation when each UAV returns to its start location is shown in Table 3. From the results, it can be concluded that the proposed relative state estimator can achieve better performance when compared with SegMap.



(a) Mapping result of proposed relative stage estimation framework aligned with Google Earth.



(b) Mapping result of proposed relative state estimation framework



(c) Mapping result of SegMap

Fig. 5 Mapping result of proposed method and SegMap.

Table 3 Translational error of relative state estimator when returning to start location.

Error	1T_2		1T_3		2T_1		2T_3		3T_1		3T_2	
	Proposed method	SegMap	Proposed method	SegMap	Proposed method	SegMap	Proposed method	SegMap	Proposed method	SegMap	Proposed method	SegMap
$x(m)$	0.09	0.54	0.22	0.15	0.01	0.72	0.10	0.65	0.02	0.38	0.22	2.41
$y(m)$	0.10	1.28	0.11	1.24	0.01	1.19	0.22	1.06	0.06	1.49	0.09	0.78
$z(m)$	0.05	2.50	0.39	2.83	0.07	0.02	0.08	2.96	0.48	0.33	0.08	2.68

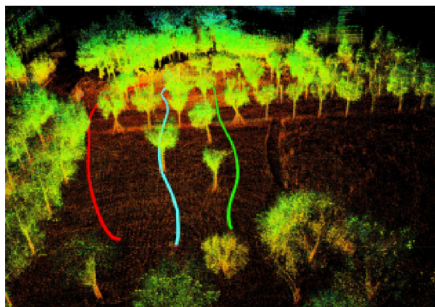
4.4. Flight test

In this test, the system consisting of 3 UAVs performs simultaneous flights in a cluttered scenario to attest to the practicality of the proposed relative state estimation framework. In the flight test, UAVs track the reference trajectories for achieving high speed while avoiding the obstacles, and the results are illustrated in Fig. 6 (the estimated trajectories of UAV₁, UAV₂, and UAV₃ are shown in blue, green, and red, respectively). During this trial, the multi-UAV system reaches a maximum speed of over 5 m/s. This experiment is carried out in a forest of approximately 45 m × 20 m, shown in Fig. 6(a). The

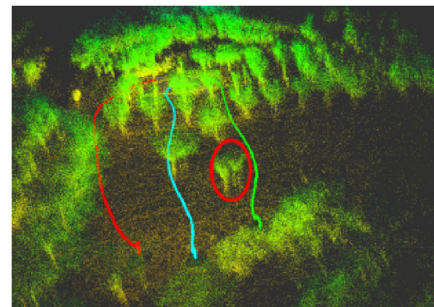
mapping and trajectories estimation results of the proposed method and SegMap are shown in Fig. 6(b) and (c), where the mapping results are merged by keyframes from 3 UAVs. Due to the inaccurate relative state estimation, the map from SegMap consists of numerous blurry structures (the 2 tree trunks that are highlighted by the red circle in Fig. 6(c) represent the same tree in reality). As shown in Fig. 6(b), the proposed method produces a map that is consistent with the real scene. To quantitatively analyze the performance of the proposed relative state estimation method, the translational error of relative state estimation when each UAV reaches its landing point is shown in Table 4. For more details, a video



(a) Scene view



(b) Mapping and trajectories estimation results of proposed method



(c) Mapping and trajectories estimation results of SegMap

Fig. 6 Real-world flight test with 3 UAVs, the multi-UAV system reaches a maximum speed of over 5 m/s.

Table 4 Translational error of relative state estimator when reaching to landing location.

Error	1T_2		1T_3		2T_1		2T_3		3T_1		3T_2	
	Proposed method	SegMap	Proposed method	SegMap	Proposed method	SegMap	Proposed method	SegMap	Proposed method	SegMap	Proposed method	SegMap
$x(m)$	0.12	0.20	0.27	0.19	0.13	0.18	0.03	0.37	0.23	0.51	0.08	0.71
$y(m)$	0.17	1.38	0.15	1.60	0.16	1.93	0.27	1.63	0.13	1.75	0.07	1.23
$z(m)$	0.04	0.34	0.11	1.28	0.03	1.82	0.12	1.30	0.09	1.83	0.06	1.74

Table 5 Average runtime for process one keyframe and communication bandwidth requirements.

Method	Average runtime for process one keyframe (ms)		Average communication bandwidth (MB)	
	Odometry	Relative state estimator	Odometry information	Keyframes
Proposed	103.58	225.13	0.004	0.093
SegMap	113.42	300.73	0.015	0.017

demonstration is available at <https://www.bilibili.com/video/BV1ef4y1q7nJ/>.

The main challenge of performing a live flight test with only the onboard computing power and communication module is to run relative state estimation in real-time with the additional workload, such as the sensor driver and the connection to the flight controller. Table 5 shows the average runtime to register one keyframe and the communication bandwidth requirements for each UAV during the flight test. As SegMap relies on external odometry measurement, a state-of-the-art lidar odometry, LOAM,¹³ is adopted during the experiment to provide odometry information. From the results, the proposed relative state estimation framework uses less runtime than SegMap, the runtime of the relative state estimator is reduced by 25%, which makes it more suitable to be deployed on a low-power onboard computer in real-time. Table 5 also compares the average communication requirements of the proposed method and SegMap. Benefit by segment approach,³⁵ SegMap provides an advantage in terms of communication burden. However, due to the segment approach provide a trade-off between communication bandwidth requirements and relative state estimation accuracy, the SegMap is less accurate than the proposed method. As shown in Table 5 the communication bandwidth requirement of the proposed method is less than 0.1 MB (the maximum communication bandwidth of the network bridge is 496 MB) during multi-UAV system aggressive flight with a maximum speed over 5 m/s.

5. Conclusions

This paper presents a distributed framework for the lidar-based relative state estimator for performing real-time trajectory estimation of the multi-UAV system in GPS-denied environments. The proposed method combines lidar-inertial odometry and loop closures together and provides a highly accurate relative state estimation result based on a factor graph. Benefiting from the keyframe strategy, each UAV performs relative state estimation individually and broadcasts very partial information, including keyframes and odometry information, without exchanging raw data. The proposed framework is validated with three experiments to demonstrate the feasibility and effectiveness. The experiment results show

that the relative estimation results reaching a precision of centimeter-level, and the proposed distributed method outperforms the centralized method SegMap, which is not limited by GPS and can be prevalently adopted for aerial swarm aggressive flight over 5 m/s.

Though the proposed method is demonstrated during a real-world simultaneous flight test with three UAVs, the 3D lidar with 360° FOV limits the take-off weight (5 kg) and wheelbase (650 mm) of our aerial platform, which makes the proposed aerial platform difficult to through a narrow flight corridor. In future work, we would like to extend the proposed method to lightweight solid-state lidar with small FOV and testing the proposed approach on different platform modalities.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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